WEEK-8 SUMMARY

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1. ASSUMPTION

There is a connection between the training and test data via a distribution. In other words, there is a probabilistic mechanism which generates the features and labels.

Every pair (x_i, y_i) is assumed to be drawn from a distribution D. Here, D is unknown, but fixed.
The connection between the training set and the set is established by D

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2. GENERATIVE MODEL

Models the following

$$\mathrm{P(x, y)}$$

Given a data point and a label, $P(x, y)$ models the probability of **(data-point,label)** being generated as a pair.

Examples :

- Naive Bayes
- Gaussian Naive Bayes

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3. DISCRIMINATIVE MODEL

Models the following

$$\mathrm{P(y|x)}$$

The end goal of binary classification is to predict whether a data point belongs to class 0 or class 1.
The above model is enough to do that.

Examples :

- DECISION TREES
  $P(y = 1|x) = 1$ if the leaf node for x is 1
- KNN
  $P(y = 0|x) = 1$ if Majority$(k - neighbours) \rightarrow$ label 0

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4. THEME

4.1. EMAIL SPAM CLASSIFICATION

$$D = \left\{(x_1, y_1), \ldots, (x_n, y_n)\right\}$$

$$x_i \in \{0, 1\}^d$$
$$y_i \in \{0, 1\}$$

Every data point corresponds to an email. Every feature corresponds to a word in some dictionary. For a data point, if the value for a feature is 1, then the corresponding word is present in the email. The value will be 0
if that word is not present

If the label of a data point is 1, then the corresponding email is a spam. If the label is 0, then the email is not a spam

4.2. GOAL

Model the following

$$P(x, y)$$

Two ways to model

$P(x, y) = P(y\|x) \times P(x)$	$P(x, y) = P(x\|y) \times P(y)$
$P(y\|x) \rightarrow$ Given that the email is generated, what is the probabiliy of the this email being a spam or what is the probabiliy of the this email not being a spam	$P(x\|y) \rightarrow$ Given that the label of an email is y, what is the probability of that email being equal to $\left[f_1, f_2, \ldots, f_d\right]$
$P(x) \rightarrow$ The probability of seeing the email x	$P(y) \rightarrow$ Distribution of labels
We just need $P(y\|x)$ for prediction. If we can come up with the structure of $P(y\|x)$, then there is no necessity to model $P(x)$	We can estimate, on average, how often the emails are spam and not spam. $P(x\|y)$ gives the distribution of emails for both the classes.

Method to be followed

$$P(x, y) = P(x|y) \times P(y)$$

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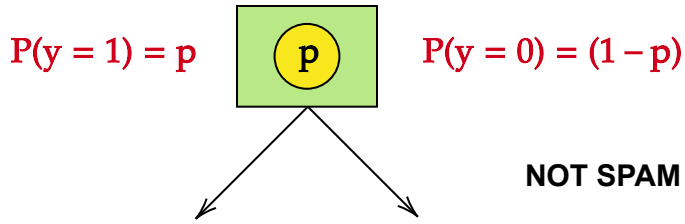
5. GENERATIVE STORY

Let us look at the story for $d = 3$ and then generalize.

$P(x = \text{"}hi,\ how\ are\ you\text{"}|1)$
$P(x = \text{"}hi,\ how\ are\ you\text{"}|0)$



SPAM

POSSIBILITES	PROBABILITIES
$[0,0,0]$	$P\left(x = [0,0,0] \middle\vert y = 1\right)$
$[0,0,1]$	$P\left(x = [0,0,1] \middle\vert y = 1\right)$
$[0,1,0]$	$P\left(x = [0,1,0] \middle\vert y = 1\right)$
$[0,1,1]$	$P\left(x = [0,1,1] \middle\vert y = 1\right)$
$[1,0,0]$	$P\left(x = [1,0,0] \middle\vert y = 1\right)$
$[1,0,1]$	$P\left(x = [1,0,1] \middle\vert y = 1\right)$
$[1,1,0]$	$P\left(x = [1,1,0] \middle\vert y = 1\right)$
$[1,1,1]$	$P\left(x = [1,1,1] \middle\vert y = 1\right)$

NOT SPAM

POSSIBILITES	PROBABILITIES
$[0,0,0]$	$P\left(x = [0,0,0] \middle\vert y = 0\right)$
$[0,0,1]$	$P\left(x = [0,0,1] \middle\vert y = 0\right)$
$[0,1,0]$	$P\left(x = [0,1,0] \middle\vert y = 0\right)$
$[0,1,1]$	$P\left(x = [0,1,1] \middle\vert y = 0\right)$
$[1,0,0]$	$P\left(x = [1,0,0] \middle\vert y = 0\right)$
$[1,0,1]$	$P\left(x = [1,0,1] \middle\vert y = 0\right)$
$[1,1,0]$	$P\left(x = [1,1,0] \middle\vert y = 0\right)$
$[1,1,1]$	$P\left(x = [1,1,1] \middle\vert y = 0\right)$

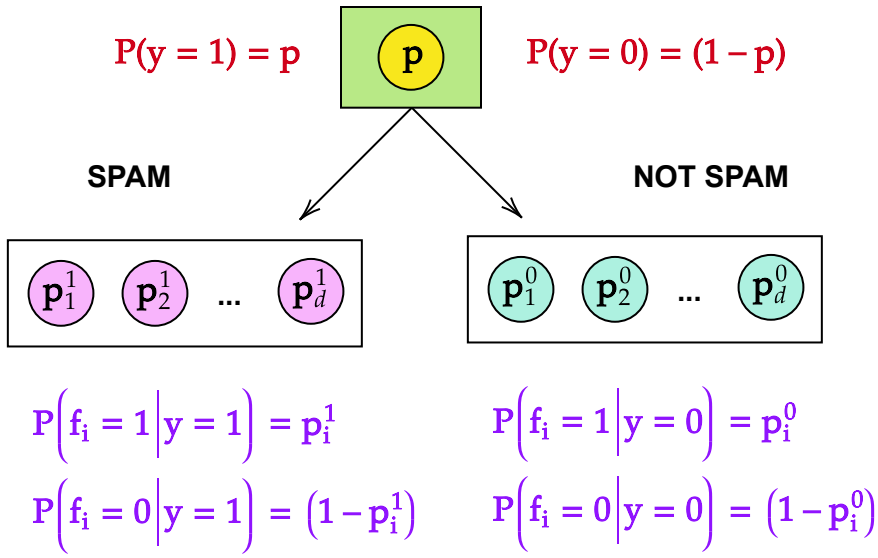
TOTAL PARAMETERS

$$2^{d+1} - 1$$

- Computationally infeasible

6. NAIVE BAYES

6.1. GENERATIVE STORY



TOTAL PARAMETERS

$$2d + 1$$

6.2. CLASS CONDITIONAL INDEPENDENCE ASSUMPTION

This model assumes the following.

$$P\left(x \middle| y\right) = P\left(x = \left[f_1, f_2, \ldots, f_d\right] \middle| y\right) = P\left(f_1 \middle| y\right) \times \ldots \times P\left(f_d \middle| y\right)$$

This assumption implies that the presence/absence of a word in an email does not have any impact on the presence or absence of any other word.

This does not model the grammatical, semantic and other structures present in a language.

However, this assumption is crucial to overcome the computational infeasibility of the previous generative story.

6.3. LIKELIHOOD FUNCTION

$$P(x, y) = P(x|y) \times P(y)$$

This can be written as

$$P(x, y) = (p)^{(y)}(1-p)^{(1-y)} \prod_{j=1}^{d} \left[\left(p_j^y\right)^{f_j} \left(1 - p_j^y\right)^{(1-f_j)} \right]$$

$$Example$$

$$(p)^{(1)}(1-p)^{(1-1)} \prod_{j=1}^{d} \left[\left(p_j^1\right)^{f_j} \left(1 - p_j^1\right)^{(1-f_j)} \right]$$

$$= p \prod_{j=1}^{d} \left(p_j^1\right)^{f_j} \left(1 - p_j^1\right)^{(1-f_j)}$$

$$(y = 1, [0, 1, 1, 0, 1, 0])$$

$$= p\left(\left(1 - p_1^1\right) \times \left(p_2^1\right) \times \left(p_3^1\right) \times \left(1 - p_4^1\right) \times \left(p_1^1\right) \times \left(1 - p_6^1\right)\right)$$

Likelihood Function

$$L(\theta; D) \propto \prod_{i=1}^{n} P(x_i, y_i)$$

This can be written as

$$L(\theta; D) \propto \prod_{i=1}^{n} (p)^{(y_i)}(1-p)^{(1-y_i)} \prod_{j=1}^{d} \left[\left(p_j^{y_i}\right)^{f_j^i} \left(1 - p_j^{y_i}\right)^{\left(1-f_j^i\right)} \right]$$

Here, f_j^i is the jth feature of the ith data point

6.4. MAXIMIZING THE LIKELIHOOD FUNCTION

Log-Likelihood

$$log(L(\theta; D)) \propto \sum_{i=1}^{n} \left[log\left(p^{(y_i)}\right) + log\left((1-p)^{(1-y_i)}\right) + log\left(\prod_{j=1}^{d} \left[\left(p_j^{y_i}\right)^{f_j^i} \left(1 - p_j^{y_i}\right)^{\left(1-f_j^i\right)} \right] \right) \right]$$

$$log(L(\theta; D)) \propto \sum_{i=1}^{n} y_i log(p) - \sum_{i=1}^{n} y_i log(1-p) + \sum_{i=1}^{n} log(1-p)$$

$$+\sum_{i=1}^{n}\sum_{j=1}^{d} f_j^i log\left(p_j^{y_i}\right) + \left(1-f_j^i\right) log\left(1-p_j^{y_i}\right)]$$

For differentiating with respect to p

$$log(p)(N1) - log(1-p)(N1) + n(log(1-p))$$

$$\frac{N1}{p} + \frac{N1}{1-p} - \frac{n}{1-p} = 0$$

$$\frac{N1}{p} = \frac{n-N1}{1-p}$$

$$N1 - pN1 = np - pN1$$

$$N1 = np$$

$$p = \frac{\sum_{i=1}^{n} y_i}{n}$$

$N1 \rightarrow$ number of data points with label being 1

$$\sum_{i=1}^{n}\sum_{j=1}^{d} f_j^i log\left(p_j^{y_i}\right) + \left(1-f_j^i\right) log\left(1-p_j^{y_i}\right) = \sum_{j=1}^{d}\sum_{i=1}^{n} f_j^i log\left(p_j^{y_i}\right) + \left(1-f_j^i\right) log\left(1-p_j^{y_i}\right)$$

$p_j^i \rightarrow p_1^1$

$$\sum_{i=1}^{n} f_1^i log\left(p_1^1\right) + \left(1-f_1^i\right) log\left(1-p_1^1\right)$$

$$log\left(p_1^1\right)\sum_{i=1}^{n} f_1^i + log\left(1-p_1^1\right)\sum_{i=1}^{n} 1 - log\left(1-p_1^1\right)\sum_{i=1}^{n} f_1^i$$

$$log\left(p_1^1\right)(NF1) + nlog\left(1-p_1^1\right) - log\left(1-p_1^1\right)(NF1)$$

$$\frac{NF1}{p_1^1} - \frac{n}{1-p_1^1} + \frac{NF1}{1-p_i^1} = 0$$

$$p_1^1 = \frac{NF1}{n}$$

$NF1$ number of data points with label being 1 and feature_1 being 1

$[1, 0, 0, 1]1$
$[1, 0, 1, 1]1$
$[0, 0, 0, 1]1$
$[1, 0, 0, 0]0$

$P(x = [1]|y = 1)$
$p_1^1 = 2/3$
$p_1^0 = 1/1$

6.5. OPTIMAL SOLUTION

$\widehat{p} \rightarrow$ Fraction of class 1 training data points

$$\widehat{p} = \frac{\sum_{i=1}^{n} y_i}{n}$$

Fraction of data points in class y training data points with the value of feature i being equal to 1)

$$p_j^y = \frac{\sum_{i=1}^{n} \mathbb{1}\left(y_i = y \;\&\; f_j^i = 1\right)}{\sum_{i=1}^{n} \mathbb{1}(y_i = y)}$$

Here, f_j^i is the jth feature of the ith data point

6.6. PREDICTION

For a test data point, the prediction is 1 if the following inequality is holds true, 0 otherwise

$$P(y_{test} = 1|x_{test}) > P(y_{test} = 0|x_{test})$$

This inequality can be written as follows.

$P(y|x) = P(y)P(x|y) / P(x)$

$$P(x_{test}, y_{test} = 1) > P(x_{test}, y_{test} = 0)$$

$$P(y_{test} = 1) \times P(x_{test}|y_{test} = 1) > P(y_{test} = 0) \times P(x_{test}|y_{test} = 0)$$

$$\widehat{p} \cdot \prod_{j=1}^{d} \left(p_j^1\right)^{f_j} \left(1 - p_j^1\right)^{(1-f_j)} > (1 - \widehat{p}) \cdot \prod_{j=1}^{d} \left(p_j^0\right)^{f_j} \left(1 - p_j^0\right)^{(1-f_j)}$$

Example

$p = 0.7,\ p_1^1 = 0.3\ , p_2^1 = 0.5\ , p_1^0 = 0.8,\ p_2^0 = 0.1$

$[0, 1] \rightarrow test$

$\left(p_2^1\right)^1 \left(1 - p_2^1\right)^{1-1}$

$P(y_t = 1|x_t) = 0.7 \times (1 - 0.3) \times (0.5)$
$P(y_t = 0|x_t) = 0.3 \times (1 - 0.8) \times (0.1)$

6.7. LAPLACE SMOOTHING

Let the value for some feature i be 0 for all the training data points.

Then, $p_i^1 = 0$ and $p_i^0 = 0$.

If the value for feature i is 1 in the test data point, then

$$\left(p_i^1\right)^{f_i} \left(1 - p_i^1\right)^{(1-f_i)} = (0)^1 (1)^0 = 0$$

$$\left(p_i^0\right)^{f_i} \left(1 - p_i^0\right)^{(1-f_i)} = (0)^1 (1)^0 = 0$$

Both $P(y_{test} = 1|x_{test})$ and $P(y_{test} = 0|x_{test})$ are product of the above two. As a result, both will become 0.

Prediction becomes difficult.
[1,1,1,1,1,1,1,....,1]→1,0 (2 emails)
We shall add two data points to our training set.

$$D = D \cup \left\{ \left([1, 1,..., 1], 1\right), \left([1, 1,..., 1], 1\right) \right\}$$

This will ensure :

$$p_i^1 \neq 0 \; and \; p_i^0 \neq 0.$$

If all the emails have a particular word, but the test data point does not have that word.

$[0, 0, 0, ..., 0] \rightarrow label\ 0,\ label\ 1$

6.8. LINEAR DECISION BOUNDARY

$$log\left(\frac{\hat{p} \cdot \prod_{j=1}^{d} \left(p_j^1\right)^{f_j} \left(1 - p_j^1\right)^{(1-f_j)}}{(1-\hat{p}) \cdot \prod_{j=1}^{d} \left(p_j^0\right)^{f_j} \left(1 - p_j^0\right)^{(1-f_j)}}\right) > log(1)$$

$$log\left(\frac{\hat{p}}{(1-\hat{p})}\right) + \sum_{j=1}^{d} f_j \left(log\left(\frac{p_j^1}{p_j^0}\right) - log\left(\frac{1-p_j^1}{1-p_j^0}\right)\right) + \sum_{j=1}^{d} log\left(\frac{1-p_j^1}{1-p_j^0}\right) > 0$$

$$c + \sum_{j=1}^{d} f_j w_j + k > 0$$

$$\sum_{j=1}^{d} f_j w_j + b > 0$$

$$[w_1, ..., w_d] \cdot [f_1, ..., f_d] + b > 0$$

$$w^T x_{test} + b > 0$$

Consider the binary classification problem with 4 binary features. The labels follow uniform$(0, 1)$ distribution. If the training dataset contains the following four points belonging to $\{0, 1\}^4$:

$[1, 0, 1, 0], [1, 1, 1, 0], [0, 0, 1, 1], [0, 1, 0, 1]$

The labels of the points are $0, 1, 1,$ and 0, respectively. Assume that the naive condition holds true.

5) With what probability does the first feature of a point take the value 0 given that the point is labeled 0?

1 point

6) Using a naive Bayes algorithm, what will be the prediction (label 0 or 1) for the point $[1, 0, 1, 0]$?

1 point

$P(y = 1) = 0.5$
$P(y = 0) = 0.5$

$2(4) + 1 \rightarrow 9$

Label 0
$[1, 0, 1, 0]$
$[0, 1, 0, 1]$

Label 1
$[1, 1, 1, 0]$
$[0, 0, 1, 1]$

$p_1^1 = \frac{1}{2}, p_2^1 = \frac{1}{2}, p_3^1 = \frac{2}{2}, p_4^1 = \frac{1}{2}$

$p_1^0 = \frac{1}{2}, p_2^0 = \frac{1}{2}, p_3^0 = \frac{1}{2}, p_4^0 = \frac{1}{2}$

$P(f_1 = 0|0) = \left(1 - p_1^0\right) = 1 - 0.5 = 0.5$

$$\widehat{p} \cdot \prod_{j=1}^{d} \left(p_j^1\right)^{f_j} \left(1 - p_j^1\right)^{(1-f_j)} > (1 - \widehat{p}) \cdot \prod_{j=1}^{d} \left(p_j^0\right)^{f_j} \left(1 - p_j^0\right)^{(1-f_j)}$$

Predict for $[1, 0, 1, 0]$

$$0.5 \times (0.5) \times (1 - 0.5) \times (1) \times (1 - 0.5) > (1 - \widehat{p}) > 0.5 \times (0.5) \times (1 - 0.5) \times (0.5) \times (1 - 0.5)$$

$$0.5^4 > 0.5^5$$

Prediction $\rightarrow$ 1

Infeasible Story

$P(y = 1), P(y = 2), \ldots, P(y = k - 1), P(y = k)$
$k \rightarrow classes$
$d \rightarrow features$

$k\left(2^d - 1\right) + (k - 1)$

$k\left(2^d\right) - k + k - 1$
$k\left(2^d\right) - 1$

$2\left(2^d - 1\right) + (2 - 1)$

$P(x, y) \rightarrow P(y) \times P(x|y)$

$p_1^0 = \frac{N0}{N0} = 1$

0
0
0
0

$p_i^0 = 0 \rightarrow p_i^1 = 1?$

$p_1^0 = 0 \rightarrow \left(1 - p_1^0\right)$